Project Idea Form

Complete one form per idea, individually. Use the nine examples in this document as models.

STUDENT NAME
Amaro, Julian Joseph

STUDENT ID
6320761

DATE
2026/09/30

PROJECT TITLE *short and descriptive*
AI Hazard Inspection Rover

THEME
☑ A ☑ B ☑ C ☐ Other

DOMAINS *at least two must be core*

Domain	Core	Supporting	Not used	How the domain is used
AI	■	☐	☐	Environmental sensor data is processed by a trained model
Robotics	■	☐	☐	mobile rover is remotely controlled and maps out an area
IoT	■	☐	☐	The watch, rover and web interface communicate wirelessly

POC QUESTION *the one uncertainty that would sink the project; it must need every core domain*

Can a remotely controlled robotic vehicle be used to sense specific gas profiles to identify and map environmental hazards?

MOTIVATION *why it matters · what is unsatisfactory today · the real engineering obstacle*

Some hazardous enviroments may be unsafe for a person to inspect directly. A remotely controlled mobile robot could allow the user to investigate an area while remaining at a safer distance. Existing basic inspection robots may provide video, but environmental sensing and automatic classification can give the operator more useful information. The main engineering challenge is combining robot control, wireless video and sensor communication, and AI-based environmental classification in one responsive system.

STAKEHOLDERS *who cares about the outcome, and why*

Stakeholder	Main concern
Emergency Responders	• Inspect potentially hazardous areas without entering directly.
	• Check if dangerous area is present.
Industrial workers	• Need to be able to see what the mobile robotic system sees.
	• Need environmental data to be transmitted remotely to check conditions
	• Need clear and easy-to-understand information to check if area is safe
	• Need real-time environmental information during an emergency

HARDWARE ALLOCATION

what each device senses and processes

Device	Sensing	Processing
Watch *ESP32-S3 AMOLED 2.06*	Touchscreen map.	Recieves the finished map and per-cell gas features over BLE. Runs the gas-profile classifier on each cell, then renders the map with color-coded hazard overlay.
Custom PCB *exactly one board*	wheel encoders, time-of-flight array gas sensor, motors and fpv camera.	contains the main microcontroller and supporting circuitry required to process sensor data, control the robot's motors, manage communication, and distribute power to the system
Computer ☐ required ■ optional ☐ not used	Touch or keyboard input	Drive page showing speed and link status. It sends velocity commands only and runs neither the real-time loop nor the model.

STACK

in execution order; tick one domain and one device per stage

#	Domain	Device	Algorithm or stage
1	☐ AI ☐ Robotics ■ IoT	☐ Watch ☐ PCB ■ Comp.	User drives robot from a phone.
2	☐ AI ■ Robotics ☐ IoT	☐ Watch ■ PCB ☐ Comp.	Robot moves safely, tracks position and builds a map of the room
3	■ AI ☐ Robotics ☐ IoT	☐ Watch ■ PCB ☐ Comp.	Gas readings are collected and tied to locations in the map
4	■ AI ☐ Robotics ☐ IoT	☐ Watch ■ PCB ☐ Comp.	Identifies what gas are present and sends to the watch.
5	☐ AI ☐ Robotics ■ IoT	■ Watch ☐ PCB ☐ Comp.	watch shows the map and location of gases

PROTOTYPE AT THE POC GATE

what must exist, and be measured, to answer the PoC question

Build *what must physically exist*

A tracked mobile rover with motor control, one custom environmental sensing/control PCB, an environmental gas sensor, camera, ESP-32 watch map, onboard AI classification, and a Wi-fi web interface displaying live video and sensor information

Out of scope *deliberately left out*

Autonomous navigation, cloud-based AI processing and production-grade hazardous-gas certification

Measure *evidence to collect*

1. Percentage of environmental test samples correctly classified by the AI model
2. Time between a watch control input and the rover responding
3. Video and environmental data update latency while the robot is being remotely controlled

Pass when *the answer is proven*

The watch map matches the room's true layout in at least 8 of 10 runs, the gases present in the room are correctly identified with 15% or less error, and the map and gas results reach the watch intact at least 90% of the time. fpv should have a minimum 1.5 seconds or less delay. Input controls should get recieved 90% of the time.

BEFORE YOU SUBMIT

- ☑ At least two domains are core, and each is used for real.
- ☑ The PoC question cannot be answered without every core domain.
- ☑ Exactly one custom PCB; remote sensors are tethered to it.
- ☑ The computer, if used, runs neither the real-time loop nor the model.
- ☑ The riskiest element can be tested in isolation.
- ☑ An ethics question (privacy, safety or bias) is identified.

INSTRUCTOR REVIEW

☐ Approved ☐ Revise

comments

initials and date